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Algebra

An Original Olympiad Practice Set

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Problems

1. DIFFICULTY 1/5 · AM-GM, WEIGHTED AM-GM, OPTIMIZATION

For $x > 0$, find the minimum value of

$$f(x) = x + \frac{4}{x^2}.$$

2. DIFFICULTY 2/5 · SUBSTITUTION, QUADRATIC IN DISGUISE, FACTORING

Find all real solutions of

$$(x^2 - 3x)^2 - 2(x^2 - 3x) = 8.$$

3. DIFFICULTY 3/5 · TELESOPING, PARTIAL FRACTIONS, SUMMATION

For a positive integer n , evaluate in closed form

$$S_n = \sum_{k=1}^n \frac{k}{(k+1)!}.$$

4. DIFFICULTY 3/5 · CAUCHY-SCHWARZ, CONCAVITY, CONSTRAINED OPTIMIZATION

Let a, b, c be positive reals with $a + b + c = 1$. Determine the maximum possible value of

$$\sqrt{a} + \sqrt{b} + \sqrt{c}.$$

5. DIFFICULTY 4/5 · SUBSTITUTION, AM-GM, MONOTONICITY TRAP

For $x > 0$, find the minimum value of

$$F(x) = x^2 + \frac{1}{x^2} + \frac{1}{x^2 + \frac{1}{x^2}}.$$

6. DIFFICULTY 4/5 · RECURRENCE, RECIPROCAL SUBSTITUTION, ARITHMETIC PROGRESSION

A sequence is defined by $a_1 = \frac{1}{2}$ and

$$a_{n+1} = \frac{a_n}{1 + 3a_n} \quad (n \geq 1).$$

Find a closed form for a_n , and evaluate $\sum_{k=1}^n a_k a_{k+1}$.

7. DIFFICULTY 5/5 · TRIGONOMETRIC SUBSTITUTION, CHEBYSHEV / TRIPLE-ANGLE, VIETA'S FORMULAS

Find all real roots of

$$8x^3 - 6x + 1 = 0,$$

expressing them in closed trigonometric form, and prove that

$$\cos 40^\circ \cos 80^\circ \cos 160^\circ = -\frac{1}{8}.$$

8. DIFFICULTY 5/5 · NEWTON'S IDENTITIES, POWER SUMS, SYMMETRIC FUNCTIONS, SYSTEMS

Real numbers x, y, z satisfy

$$x + y + z = 2, \quad x^2 + y^2 + z^2 = 6, \quad x^3 + y^3 + z^3 = 8.$$

Find the unordered triple $\{x, y, z\}$ and the value of $x^4 + y^4 + z^4$.

Solutions

1. Answer: 3

The trick is to split x into two equal pieces so that AM-GM can be applied to three terms whose product is a constant. Write

$$f(x) = \frac{x}{2} + \frac{x}{2} + \frac{4}{x^2}.$$

These three positive numbers have product

$$\frac{x}{2} \cdot \frac{x}{2} \cdot \frac{4}{x^2} = 1,$$

which is independent of x . By AM-GM,

$$\frac{x}{2} + \frac{x}{2} + \frac{4}{x^2} \geq 3\sqrt[3]{\frac{x}{2} \cdot \frac{x}{2} \cdot \frac{4}{x^2}} = 3\sqrt[3]{1} = 3.$$

Equality requires all three terms equal: $\frac{x}{2} = \frac{4}{x^2}$, i.e. $x^3 = 8$, so $x = 2$. Then $f(2) = 2 + \frac{4}{4} = 3$. Hence $\min f = 3$.

Second method. Calculus: $f'(x) = 1 - \frac{8}{x^3} = 0$ gives $x^3 = 8$, so $x = 2$. Since $f''(x) = \frac{24}{x^4} > 0$, this is a minimum, and $f(2) = 3$. The point of the AM-GM method is that splitting x into the right number of equal pieces is exactly what makes the geometric mean constant.

2. Answer: $x \in \{-1, 1, 2, 4\}$

Set $u = x^2 - 3x$. The equation becomes

$$u^2 - 2u - 8 = 0 \implies (u - 4)(u + 2) = 0,$$

so $u = 4$ or $u = -2$. Now resolve each back in x . For $u = 4$: $x^2 - 3x - 4 = 0 \Rightarrow (x - 4)(x + 1) = 0 \Rightarrow x = 4$ or $x = -1$. For $u = -2$: $x^2 - 3x + 2 = 0 \Rightarrow (x - 1)(x - 2) = 0 \Rightarrow x = 1$ or $x = 2$. All four values are real, so the solution set is $\{-1, 1, 2, 4\}$.

Second method. Direct expansion gives the quartic $x^4 - 6x^3 + 7x^2 + 6x - 8 = 0$, which factors as $(x + 1)(x - 1)(x - 2)(x - 4) = 0$, recovering the same four roots. The substitution simply reveals this factorization without brute-force quartic work.

3. Answer: $S_n = 1 - \frac{1}{(n+1)!}$

The key is to write the numerator as a difference that telescopes against the factorial. Since $k = (k+1) - 1$,

$$\frac{k}{(k+1)!} = \frac{(k+1) - 1}{(k+1)!} = \frac{k+1}{(k+1)!} - \frac{1}{(k+1)!} = \frac{1}{k!} - \frac{1}{(k+1)!}.$$

Therefore the sum telescopes:

$$S_n = \sum_{k=1}^n \left(\frac{1}{k!} - \frac{1}{(k+1)!} \right) = \frac{1}{1!} - \frac{1}{(n+1)!} = 1 - \frac{1}{(n+1)!}.$$

As a sanity check, $n = 1$ gives $\frac{1}{2} = 1 - \frac{1}{2}$; $n = 2$ gives $\frac{1}{2} + \frac{2}{6} = \frac{5}{6} = 1 - \frac{1}{6}$.

Second method. Guess-and-induct: from small cases $S_1 = \frac{1}{2}$, $S_2 = \frac{5}{6}$, $S_3 = \frac{23}{24}$ one conjectures $S_n = 1 - \frac{1}{(n+1)!}$. For the inductive step, $S_{n+1} = S_n + \frac{n+1}{(n+2)!} = 1 - \frac{1}{(n+1)!} + \frac{n+1}{(n+2)!} = 1 - \frac{(n+2)-(n+1)}{(n+2)!} = 1 - \frac{1}{(n+2)!}$, completing the induction.

4. Answer: $\sqrt{3}$

By the Cauchy-Schwarz inequality applied to the vectors $(\sqrt{a}, \sqrt{b}, \sqrt{c})$ and $(1, 1, 1)$,

$$(\sqrt{a} + \sqrt{b} + \sqrt{c})^2 \leq (a + b + c)(1^2 + 1^2 + 1^2) = 1 \cdot 3 = 3.$$

Hence $\sqrt{a} + \sqrt{b} + \sqrt{c} \leq \sqrt{3}$. Equality in Cauchy-Schwarz holds when $(\sqrt{a}, \sqrt{b}, \sqrt{c})$ is proportional to $(1, 1, 1)$, i.e. $a = b = c = \frac{1}{3}$; then the sum is $3\sqrt{1/3} = \sqrt{3}$. So the maximum is $\sqrt{3}$. (Note there is no minimum over the open region: pushing two variables toward 0 drives the sum toward 1, but 1 is never attained.)

Second method. Jensen / concavity: $g(t) = \sqrt{t}$ is concave on $[0, \infty)$, so

$$\frac{\sqrt{a} + \sqrt{b} + \sqrt{c}}{3} \leq \sqrt{\frac{a+b+c}{3}} = \sqrt{\frac{1}{3}},$$

giving $\sqrt{a} + \sqrt{b} + \sqrt{c} \leq 3\sqrt{1/3} = \sqrt{3}$, with equality at $a = b = c = \frac{1}{3}$.

5. Answer: $\frac{5}{2}$

Substitute $t = x^2 + \frac{1}{x^2}$. By AM-GM, $t = x^2 + \frac{1}{x^2} \geq 2$, with equality iff $x^2 = \frac{1}{x^2}$, i.e. $x = 1$. The expression becomes

$$F = g(t) = t + \frac{1}{t}, \quad t \geq 2.$$

Here is the trap: the unconstrained minimum of $g(t) = t + \frac{1}{t}$ is at $t = 1$, but $t = 1$ is not attainable since $t \geq 2$. On $[2, \infty)$ we have $g'(t) = 1 - \frac{1}{t^2} > 0$, so g is strictly increasing there; its minimum is therefore at the left endpoint $t = 2$:

$$g(2) = 2 + \frac{1}{2} = \frac{5}{2}.$$

This is achieved exactly when $t = 2$, i.e. $x = 1$. Hence $\min F = \frac{5}{2}$.

Second method. Direct AM-GM is invalid here because applying $t + \frac{1}{t} \geq 2$ would suggest a minimum of 2, which is unattainable given $t \geq 2$; recognizing this is the whole point. To confirm with calculus, at $x = 1$, $F(1) = 1 + 1 + \frac{1}{2} = \frac{5}{2}$, and a check of $F'(x)$ shows $x = 1$ is the only interior critical point and yields the global minimum on $(0, \infty)$ (with $F \rightarrow \infty$ as $x \rightarrow 0^+$ or $x \rightarrow \infty$).

6. Answer: $a_n = \frac{1}{3n-1}$; $\sum_{k=1}^n a_k a_{k+1} = \frac{1}{3} \left(\frac{1}{2} - \frac{1}{3n+2} \right) = \frac{n}{2(3n+2)}$

Each term is positive, so we may take reciprocals. Let $b_n = \frac{1}{a_n}$. Then

$$b_{n+1} = \frac{1 + 3a_n}{a_n} = \frac{1}{a_n} + 3 = b_n + 3.$$

Thus (b_n) is an arithmetic progression with first term $b_1 = \frac{1}{a_1} = 2$ and common difference 3: $b_n = 2 + 3(n-1) = 3n-1$. Hence

$$a_n = \frac{1}{3n-1}.$$

For the sum, use partial fractions:

$$a_k a_{k+1} = \frac{1}{(3k-1)(3k+2)} = \frac{1}{3} \left(\frac{1}{3k-1} - \frac{1}{3k+2} \right).$$

This telescopes:

$$\sum_{k=1}^n a_k a_{k+1} = \frac{1}{3} \left(\frac{1}{2} - \frac{1}{3n+2} \right) = \frac{1}{3} \cdot \frac{3n}{2(3n+2)} = \frac{n}{2(3n+2)}.$$

Second method. One can verify the closed form by induction: it holds at $n = 1$ since $a_1 = \frac{1}{2} = \frac{1}{3 \cdot 1 - 1}$, and if $a_n = \frac{1}{3n-1}$ then $a_{n+1} = \frac{1/(3n-1)}{1+3/(3n-1)} = \frac{1}{(3n-1)+3} = \frac{1}{3(n+1)-1}$. The reciprocal substitution is what reveals this form in the first place, turning a nonlinear recurrence into a linear (arithmetic) one.

7. Answer: Roots: $x = \cos 40^\circ, \cos 80^\circ, \cos 160^\circ$; **and the product equals $-\frac{1}{8}$.** (=Proof for the identity.)

Substitute $x = \cos \theta$. Using the triple-angle identity $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$, we have $4x^3 - 3x = \cos 3\theta$. Divide the equation by 2:

$$4x^3 - 3x = -\frac{1}{2} \implies \cos 3\theta = -\frac{1}{2}.$$

Thus $3\theta = 120^\circ + 360^\circ k$ or $3\theta = 240^\circ + 360^\circ k$, giving $\theta \in \{40^\circ, 80^\circ, 160^\circ\}$ (the three values in $[0^\circ, 180^\circ]$ yielding distinct cosines). Hence the three real roots are

$$x = \cos 40^\circ, \cos 80^\circ, \cos 160^\circ,$$

and since a cubic has at most three roots, these are all of them. Now apply Vieta's formulas to $8x^3 + 0 \cdot x^2 - 6x + 1 = 0$: the product of the roots is $-\frac{\text{constant}}{\text{leading}} = -\frac{1}{8}$. Therefore

$$\cos 40^\circ \cos 80^\circ \cos 160^\circ = -\frac{1}{8}.$$

Second method. Independent proof of the product identity via the doubling telescope. Let $P = \cos 40^\circ \cos 80^\circ \cos 160^\circ$. Multiply by $\sin 40^\circ$ and use $2 \sin \alpha \cos \alpha = \sin 2\alpha$ repeatedly: $8 \sin 40^\circ P = 8 \sin 40^\circ \cos 40^\circ \cos 80^\circ \cos 160^\circ = 4 \sin 80^\circ \cos 80^\circ \cos 160^\circ = 2 \sin 160^\circ \cos 160^\circ = \sin 320^\circ = -\sin 40^\circ$. Since $\sin 40^\circ \neq 0$, $8P = -1$, so $P = -\frac{1}{8}$, confirming the Vieta computation.

8. Answer: $\{x, y, z\} = \{-1, 1, 2\}$ and $x^4 + y^4 + z^4 = 18$

Let $p_k = x^k + y^k + z^k$ and let e_1, e_2, e_3 be the elementary symmetric polynomials, so x, y, z are the roots of

$$t^3 - e_1 t^2 + e_2 t - e_3 = 0.$$

We are given $p_1 = 2, p_2 = 6, p_3 = 8$. \emph{Newton's identities} give e_1, e_2, e_3 :

$$e_1 = p_1 = 2,$$

$$e_2 = \frac{e_1 p_1 - p_2}{2} = \frac{2 \cdot 2 - 6}{2} = -1,$$

$$e_3 = \frac{e_2 p_1 - e_1 p_2 + p_3}{3} = \frac{(-1)(2) - (2)(6) + 8}{3} = \frac{-2 - 12 + 8}{3} = -2.$$

Hence the cubic is

$$t^3 - 2t^2 - t + 2 = 0.$$

Factor by grouping: $t^2(t - 2) - (t - 2) = (t - 2)(t^2 - 1) = (t - 2)(t - 1)(t + 1) = 0$, so the roots are $t = -1, 1, 2$. Therefore $\{x, y, z\} = \{-1, 1, 2\}$. Finally, Newton's identity for p_4 (with three variables, $e_4 = 0$) reads

$$p_4 = e_1 p_3 - e_2 p_2 + e_3 p_1 = 2 \cdot 8 - (-1) \cdot 6 + (-2) \cdot 2 = 16 + 6 - 4 = 18.$$

Directly, $(-1)^4 + 1^4 + 2^4 = 1 + 1 + 16 = 18$, confirming $x^4 + y^4 + z^4 = 18$.

Second method. Avoid Newton's identities by hand: $e_2 = \frac{1}{2}[(x + y + z)^2 - (x^2 + y^2 + z^2)] = \frac{1}{2}(4 - 6) = -1$. For e_3 , use the identity $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$: since $xy + yz + zx = e_2 = -1$, the right side is $2(6 - (-1)) = 14$, so $8 - 3e_3 = 14 \Rightarrow e_3 = -2$. This reproduces the cubic $t^3 - 2t^2 - t + 2 = 0$ and the roots $\{-1, 1, 2\}$, after which $x^4 + y^4 + z^4 = 18$ follows by direct substitution.